

Reflection Log

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Credit Name: CSE3910 - Project D

Assignment Name: AvoidObstacles

How has your program changed from planning to coding to now?

Initially, this program was designed to make the rover navigate autonomously using the distance sensor to detect and avoid obstacles. While the primary objective was achieved, unexpected challenges arose during development, leading to adjustments that enhanced the program's reliability and performance.

1.Distance Measurement Delays

At first, the program experienced issues when processing rapid distance measurements, which caused erratic movements. To address this, I fine-tuned the data interval of the sensor to a manageable value, ensuring smoother responses.

Code Example:

```
sonar.setDataInterval(100); // Set a manageable interval for readings
```

2. Reversing Logic Refinement

During testing, the rover would sometimes fail to back up effectively before turning. I adjusted the reversing logic to ensure enough time for the rover to clear the obstacle.

Code Example:

```
leftMotors.setTargetVelocity(-0.5);  
rightMotors.setTargetVelocity(-0.5);  
Thread.sleep(700); // Increased duration for better clearance
```

3.Turn Consistency Issues

Uneven ground conditions caused inconsistent turns. I addressed this by normalizing motor speeds and re-checking distances after turning.

Code Example:

```
leftMotors.setTargetVelocity(-0.5);  
rightMotors.setTargetVelocity(0.5);  
Thread.sleep(500); // Adjusted timing for accurate turns
```

4. Resource Management

I noticed the program didn't handle closing resources like motors and sensors properly, so I added cleanup logic at the end.

Code Example:

```
leftMotors.close();  
rightMotors.close();  
sonar.close();
```

Final Thoughts: These changes made the program more predictable and efficient. The rover can now consistently avoid obstacles without erratic behavior, creating a smoother and more reliable experience.